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Course: CyberOps
Lab: Linux Servers
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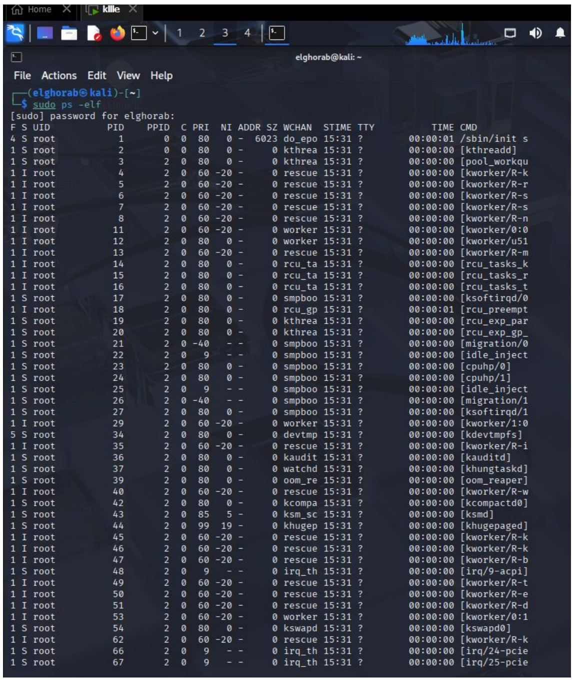
Screenshot 1 – Displaying Running Processes



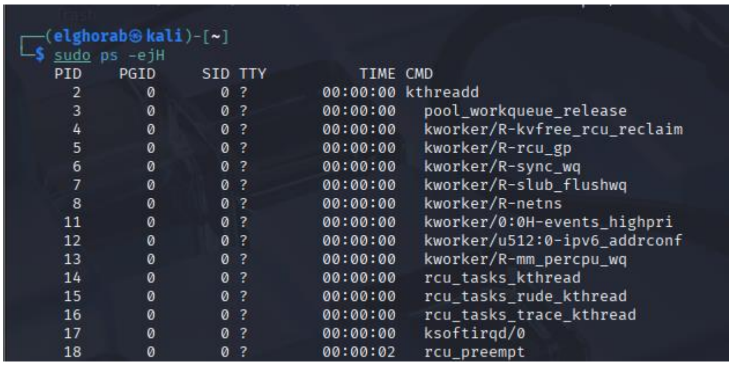
The `sudo ps -elf` command was used to display all running processes on the Linux system.

Q:1

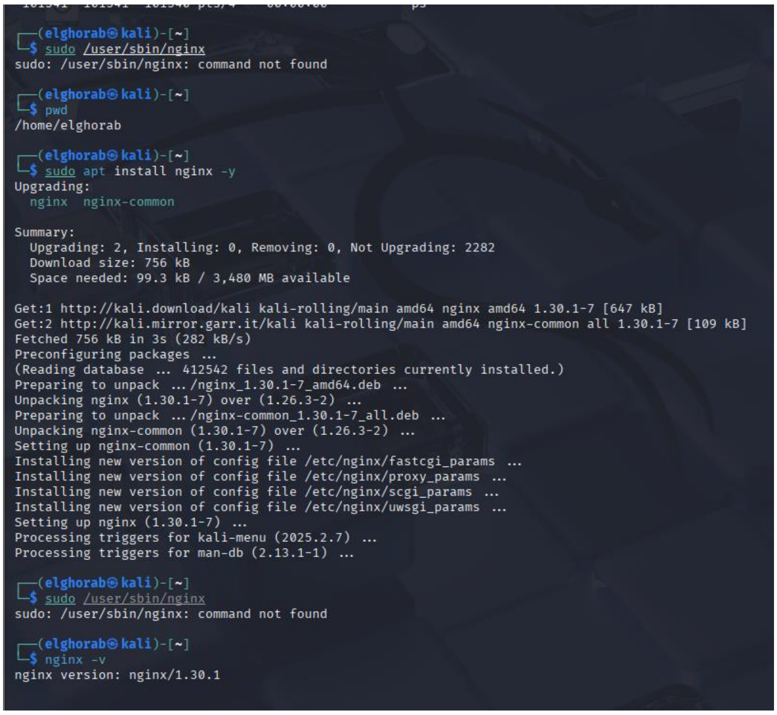
Why was it necessary to run **ps** as root (prefacing the command with **sudo**)?

It was necessary to run `ps` as root because some processes are owned by other users or the root user. Using `sudo` allows us to view complete

Screenshot 2 – Displaying Running Processes

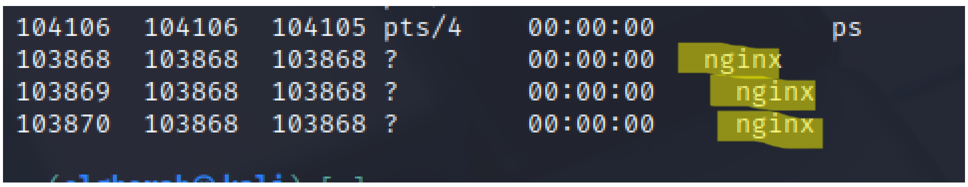


Note: Nginx was not installed on the system initially, so I installed it manually before continuing with the lab steps.



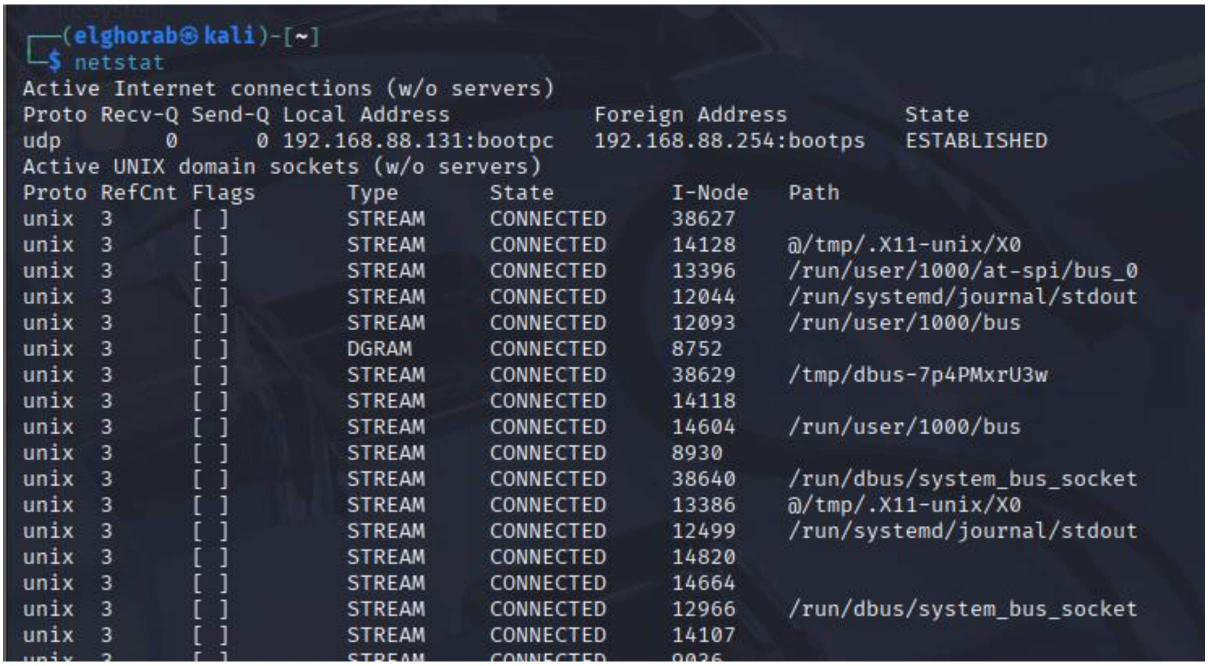
First, I installed **Nginx** using the `apt` package manager. After the installation was completed successfully, I checked the installed Nginx version using the `nginx -v` command. The installed version was **Nginx 1.30.1**.

Then, I started the Nginx web server and displayed its information. The commands were executed successfully, and the output is shown in the screenshot below.



Q:2:- **How is the process hierarchy represented by ps**

The process hierarchy is represented by showing parent and child processes. The child process is indented under its parent process.



```
┌──(elghorab㉿kali)-[~]
└─$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 192.168.88.131:bootpc   192.168.88.254:bootps   ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     38627
unix  3      [ ]         STREAM     CONNECTED     14128    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     13396    /run/user/1000/at-spi/bus_0
unix  3      [ ]         STREAM     CONNECTED     12044    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     12093    /run/user/1000/bus
unix  3      [ ]         DGRAM      CONNECTED     8752
unix  3      [ ]         STREAM     CONNECTED     38629    /tmp/dbus-7p4PMxrU3w
unix  3      [ ]         STREAM     CONNECTED     14118
unix  3      [ ]         STREAM     CONNECTED     14604    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     8930
unix  3      [ ]         STREAM     CONNECTED     38640    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     13386    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     12499    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     14820
unix  3      [ ]         STREAM     CONNECTED     14664
unix  3      [ ]         STREAM     CONNECTED     12966    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     14107
```

```
└─$ sudo netstat -tunap
[sudo] password for elghorab:
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 192.168.88.131:8080     0.0.0.0:*               LISTEN      79367/python3
tcp        0      0 0.0.0.0:80              0.0.0.0:*               LISTEN      103868/nginx: maste
tcp6       0      0 :::80                   :::*                    LISTEN      103868/nginx: maste
udp        0      0 192.168.88.131:68       192.168.88.254:67       ESTABLISHED 737/NetworkManager
```

Q3:- What is the meaning of the **–t**, **-u**, **–n**, **–a** and **–p** options in **netstat**? (use **man netstat** to answer)
-t --> TCP connections
-u--> UDP connections
-n--> Show numerical addresses/ports
-a--> Show all connections and listening ports
-p--> Show PID/program name

Q4:- **Is the order of the options important to netstat?**

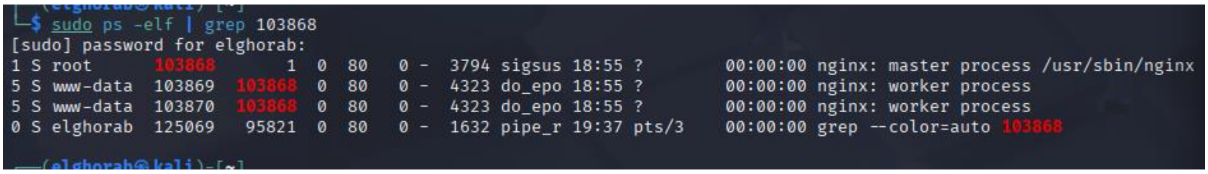
No. The order of these options is not important as long as they are provided as options to netstat

Based on the netstat output shown in item (d), what is the Layer 4 protocol, connection status, and PID of the process running on port 80?

Protocol: TCP
Status: LISTEN
PID: 103868

Q:4:-What kind of service is running on port 80 TCP?

HTTP web server.



```
└─$ sudo ps -elf | grep 103868
[sudo] password for elghorab:
1 S root      103868      1  0  80   0 -  3794 sigsus 18:55 ?        00:00:00 nginx: master process /usr/sbin/nginx
5 S www-data  103869 103868  0  80   0 -  4323 do_epo 18:55 ?        00:00:00 nginx: worker process
5 S www-data  103870 103868  0  80   0 -  4323 do_epo 18:55 ?        00:00:00 nginx: worker process
0 S elghorab  125069  95821  0  80   0 -  1632 pipe_r 19:37 pts/3    00:00:00 grep --color=auto 103868
```

Q5:- How could it be concluded that the process PID is nginx?

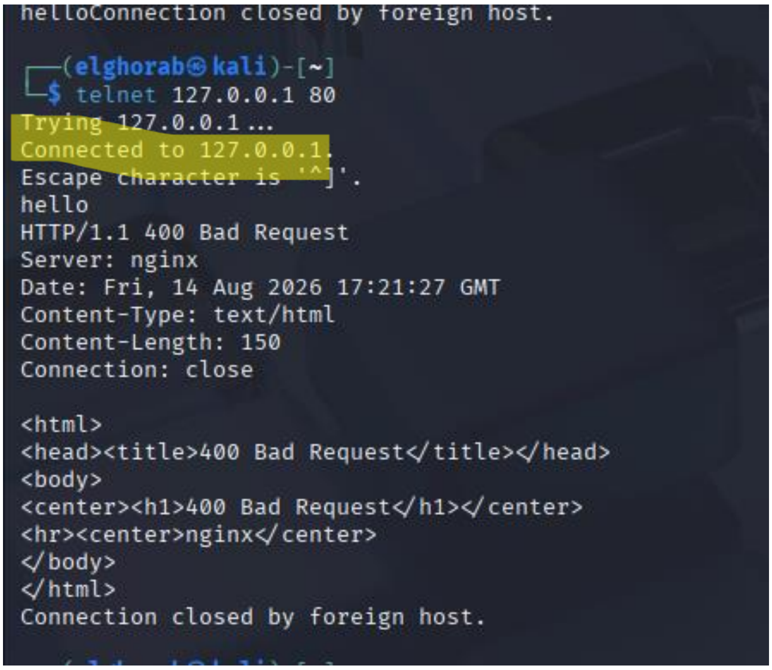
The ps output shows the nginx process name and command associated with that PID, including nginx master process

Q6:-What does it mean that process 396 has process 395 as its parent?

It means that process 395 is the parent process that started process 396. This is common because servers such as nginx commonly use a master process that creates worker processes.

Q7:- Why is the last line showing `grep 395`?

Because the grep command itself contains the number 395, so the ps output also matches the grep search.



```
helloConnection closed by foreign host.

┌──(elghorab㉿kali)-[~]
└─$ telnet 127.0.0.1 80
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^]'.
hello
HTTP/1.1 400 Bad Request
Server: nginx
Date: Fri, 14 Aug 2026 17:21:27 GMT
Content-Type: text/html
Content-Length: 150
Connection: close

<html>
<head><title>400 Bad Request</title></head>
<body>
<center><h1>400 Bad Request</h1></center>
<hr><center>nginx</center>
</body>
</html>
Connection closed by foreign host.
```

Q8:- Why was the error sent as a web page

Because nginx is a web server and it responds to HTTP requests using HTTP responses formatted as web pages.

```
┌──(elghorab㉿kali)-[~]
└─$ telnet 127.0.0.1 22
Trying 127.0.0.1 ...
telnet: Unable to connect to remote host: Connection refused

┌──(elghorab㉿kali)-[~]
```

Telnet Test on Port 22:

I attempted to connect to the local host on TCP port 22 using Telnet. The connection was refused, indicating that no service was listening on port 22 at the time of the test.

```
┌──(elghorab㉿kali)-[~]
└─$ telnet 127.0.0.1 68
Trying 127.0.0.1 ...
telnet: Unable to connect to remote host: Connection refused
```

Telnet Test on Port 68

I attempted to connect to the local host on TCP port 68 using Telnet. The connection was refused, indicating that no service was listening on port 68 at the time of the test.

1. What are the advantages of using netstat?

Netstat can display active network connections, listening ports, protocols, local and remote addresses, connection states, and the PID/program associated with network services.

2. What are the advantages of using Telnet? Is it safe?

Telnet can be useful for testing TCP services and identifying the type of service running on a specific port. However, Telnet is not safe for remote administration because it does not encrypt session data and can expose